

Week 8 Types of Model and Naive Bayes Algorithm

Jan 2024 Term - Quiz 2

Question 8 : 640653770622

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider a naive Bayes model is trained on the following data matrix X of shape (d, n) and corresponding label vector y :

$$X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \quad y = [1 \quad 0 \quad 1 \quad 1]^T$$

Assume that $\hat{p}$ and $\hat{p}_j^{y_i}$ are estimates for $P(y = 1)$ and $P(f_j = 1|y = y_i)$, respectively. Here, f_i ; $i = 1, 2, 3$ is the i^{th} feature.

These parameters are estimated using MLE.

Question 9 : 640653770623

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : SA

Calculate the value of $\hat{p}_2^1$

Answer (Numeric):

Answer

Solution

$$\hat{p}_j^y = \frac{\sum_{i=1}^n \mathbb{1}(f_j^i = 1, y_i = y)}{\sum_{i=1}^n \mathbb{1}(y_i = y)}$$

$$\hat{p}_2^1 = \frac{\sum_{i=1}^4 \mathbb{1}(f_2^i = 1, y_i = 1)}{\sum_{i=1}^4 \mathbb{1}(y_i = 1)} = \frac{0 + 0 + 0 + 1}{1 + 0 + 1 + 1} = \frac{1}{3}$$

Question 10 : 640653770624

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : SA

Calculate the value of $\hat{p}_1^1$

Answer (Numeric):

Answer

Solution

$$\hat{p}_1^1 = \frac{\sum_{i=1}^4 \mathbb{1}(f_1^i = 1, y_i = 1)}{\sum_{i=1}^4 \mathbb{1}(y_i = 1)} = \frac{1 + 0 + 0 + 0}{1 + 0 + 1 + 1} = \frac{1}{3}$$

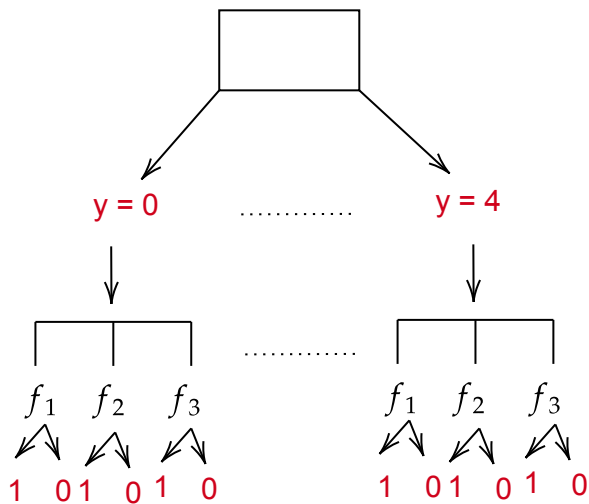
Suppose you have a five-class classification problem where class label $y \in (0, 1, 2, 3, 4)$ and each training example x_i has binary features $f_1, f_2, f_3 \in (0, 1)$. How many parameters do we need to know to classify an example using Naive Bayes classifier?

Answer (Numeric):

Answer

Solution

Decide labels = $5 - 1 = 4$
Estimate each features under each label = $2 - 1 = 1$
Estimate all features under each label = $1 + 1 + 1 = 3$
Estimate all features under all labels is $3 + 3 + 3 + 3 + 3 = 15$
Total parameters we need to know is $15 + 4 = 19$



Assume that there is no smoothing. Answer the given subquestions.

Consider the following dataset for a binary classification problem in which the features are from $\{0,1\}^3$ and the labels are from $\{0,1\}$.

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$
$$\mathbf{y} = [1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0]^T$$

A Naive Bayes classifier is trained on this dataset. The parameters to be estimated are represented as $\hat{p}_i^y$, which are presented in the form of the table given below. Recall that $\hat{p}_i^y$ is the probability of feature i taking a value 1 in class y :

i	$y = 0$	$y = 1$
1	a	b
2	c	d
3	e	f

Solution

$$\hat{p}_{ml} = \frac{\sum_{i=1}^n y_i}{n} = \frac{4}{8} = \frac{1}{2}$$

$$\hat{p}_j^y = \frac{\sum_{i=1}^n \mathbb{1}(f_j^i = 1, y_i = y)}{\sum_{i=1}^n \mathbb{1}(y_i = y)}$$

i	$y = 0$	$y = 1$
1	$\frac{1}{4}$	$\frac{3}{4}$
2	$\frac{3}{4}$	$\frac{2}{4}$
3	$\frac{2}{4}$	$\frac{3}{4}$

Find the value of c + d.

Answer (Numeric):

Answer

Solution

$$\frac{3}{4} + \frac{2}{4} = \frac{5}{4} = 1.25$$

Question 26 : 640653852847

 View Parent QN

 View Solutions (0)

Total Mark : 2.00 | Type : SA

Find the value of a + c + e.

Answer (Numeric):

Answer

Solution

$$\frac{1}{4} + \frac{3}{4} + \frac{2}{4} = 1.50$$

Question 27 : 640653852848

 View Parent QN

 View Solutions (0)

Total Mark : 2.00 | Type : SA

Find the predicted label for the 

test point $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$.

Answer (Numeric):

Answer

Solution

$$\begin{aligned} P(\hat{y}_{test} = 1 \mid x_{test}) &> P(\hat{y}_{test} = 0 \mid x_{test}) \\ \implies \hat{y}_{test} &= 1 \\ &0, \text{ otherwise} \end{aligned}$$

$$\begin{aligned} P(\hat{y}_{test} = 1 \mid x_{test}) &= P(x_{test} \mid \hat{y}_{test} = 1) P(\hat{y}_{test} = 1) \\ P(x_{test} = [0 \ 1 \ 0] \mid \hat{y}_{test} = 1) &P(\hat{y}_{test} = 1) \\ &= \left(\prod_{j=1}^d (\hat{p}_j^{f_j} (1 - \hat{p}_j)^{(1-f_j)}) \right) \hat{p} \\ &= \left(\frac{1}{4} \times \frac{2}{4} \times \frac{1}{4} \right) \times \frac{1}{2} \\ &= 0.015625 \end{aligned}$$

$$\begin{aligned}
 P(\hat{y}_{test} = 0 \mid x_{test}) &= P(x_{test} \mid \hat{y}_{test} = 0) P(\hat{y}_{test} = 0) \\
 P(x_{test} = [0 \ 1 \ 0] \mid \hat{y}_{test} = 0) P(\hat{y}_{test} = 0) \\
 &= \left(\prod_{j=1}^d (\hat{p}_j^0)^{f_j} (1 - \hat{p}_j^0)^{(1-f_j)} \right) (1 - \hat{p}) \\
 &= \left(\frac{3}{4} \times \frac{1}{4} \times \frac{2}{4} \right) \times \frac{1}{2} \\
 &= 0.046875
 \end{aligned}$$

$$\begin{aligned}
 &\because 0.046875 > 0.015625 \\
 \implies P(\hat{y}_{test} = 0 \mid x_{test}) &> P(\hat{y}_{test} = 1 \mid x_{test}) \\
 \implies \hat{y}_{test} &= 0
 \end{aligned}$$

Question 28 : 640653852849

Total Mark : 0.00 | Type : COMPREHENSION

Consider a multi-class classification problem with 3 classes and 4 features, all of which are binary. A generative model is used to model the joint distribution of the features and labels. Based on the above data, answer the given subquestions.

Question 29 : 640653852850

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

Find the total number of independent (free) parameters in the model if the class conditional independence assumption is not enforced.

Answer (Numeric):

Answer

Solution

$$\begin{aligned}
 \text{Number of parameters} &= (k - 1) + (2^d - 1) + \dots + (2^d - 1) \\
 &= k - 1 + k(2^d) - k \\
 &= k(2^d) - 1 \\
 &= 3(2^4) - 1 \\
 &= 47
 \end{aligned}$$

Find the total number of independent (free) parameters in the model if the class conditional independence assumption is enforced.

Answer (Numeric):

Solution

Decide labels = $3 - 1 = 2$

Estimate each features under each label = $2 - 1 = 1$

Estimate all features under each label = $1 + 1 + 1 + 1 = 4$

Estimate all features under all labels is $4 + 4 + 4 = 12$

Total parameters we need to know is $2 + 12 = 14$

Based on the above data, answer the given subquestions.

A binary classification dataset has 2000 data points belonging to $\{0, 1\}^2$.
A Naive Bayes algorithm was run on the same dataset, resulting in the following estimates:

- $\hat{p}$, estimate for $P(y = 1) = 0.4$
- $\hat{p}_1^0$, estimate for $P(f_1 = 1 \mid y = 0) = 0.25$
- $\hat{p}_2^0$, estimate for $P(f_2 = 1 \mid y = 0) = 0.35$
- $\hat{p}_1^1$, estimate for $P(f_1 = 1 \mid y = 1) = 0.15$
- $\hat{p}_2^1$, estimate for $P(f_2 = 1 \mid y = 1) = 0.05$

Question 6 : 6406531021046

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 4.00 | Type : SA

What is the estimated value of $P(f_2 = 0 \mid y = 1)$?
Write your answer correct to two decimal places.

Answer (Numeric):

Answer

Solution

$$P(f_2 = 0|y = 1) = 1 - 0.05 = 0.95$$

Question 7 : 6406531021047

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : SA

What will be the predicted label for the data point $[1, 0]$?

Answer (Numeric):

Answer

Solution

$$\begin{aligned} P(\hat{y}_{test} = 1 \mid x_{test}) &> P(\hat{y}_{test} = 0 \mid x_{test}) \\ \implies \hat{y}_{test} &= 1 \\ &0, \text{ otherwise} \end{aligned}$$
$$\begin{aligned} P(\hat{y}_{test} = 1 \mid x_{test}) &= P(x_{test} \mid \hat{y}_{test} = 1) P(\hat{y}_{test} = 1) \\ &= P\left(x_{test} = \begin{bmatrix} 1 & 0 \end{bmatrix} \mid \hat{y}_{test} = 1\right) P(\hat{y}_{test} = 1) \\ &= \left(\prod_{j=1}^d \left(\hat{p}_j^1\right)^{f_j} \left(1 - \hat{p}_j^1\right)^{(1-f_j)}\right) \hat{p} \\ &= (0.15 \times 0.95) \times 0.4 \\ &= 0.057 \end{aligned}$$

$$\begin{aligned}
 P(\hat{y}_{test} = 0 \mid x_{test}) &= P(x_{test} \mid \hat{y}_{test} = 0) P(\hat{y}_{test} = 0) \\
 &= P(x_{test} = [1 \ 0] \mid \hat{y}_{test} = 0) P(\hat{y}_{test} = 0) \\
 &= \left(\prod_{j=1}^d (\hat{p}_j^0)^{f_j} (1 - \hat{p}_j^0)^{(1-f_j)} \right) (1 - \hat{p}) \\
 &= (0.25 \times 0.65) \times 0.6 \\
 &= 0.0975
 \end{aligned}$$

$$\begin{aligned}
 &\because 0.0975 > 0.057 \\
 \Rightarrow P(\hat{y}_{test} = 0 \mid x_{test}) &> P(\hat{y}_{test} = 1 \mid x_{test}) \\
 \Rightarrow \hat{y}_{test} &= 0
 \end{aligned}$$

Question 10 : 6406531021044

[View Solutions \(0\)](#)

Total Mark : 4.00 | Type : SA

Suppose you want to use a Naive Bayes classifier to predict whether a student will pass or fail an exam based on two features: the number of hours they studied and whether they attended review sessions. Assume that the features are conditionally independent given the exam outcome and that the variances of the study hours distributions are equal for both pass and fail categories.

How many parameters are required to classify a new student using this Naive Bayes classifier?

Answer (Numeric):

Answer

Solution

Decide label = 2 - 1 = 1

Parameter of feature $f_1 = \mu_1, \mu_2, \Sigma$

Parameter of feature f_2 for each label = 2 - 1 = 1

Parameter of feature f_2 for all label = 1 + 1 = 2

Total parameters we need to know is 1 + 3 + 2 = 6